

Bookish

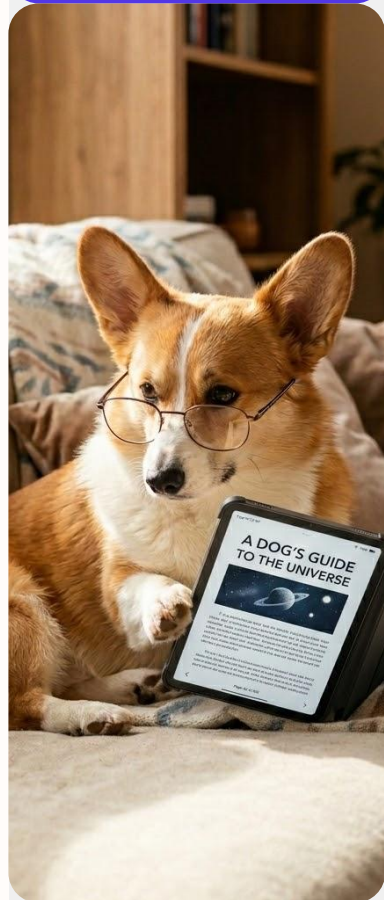
Project 2: AI Modeling in Practice

A collaborative filtering and personalization model for book recommendations

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AI Use Disclaimer: Claude Code (development), Gemini AI (LLM re-ranking and slide design enhancements)



Overview: Problem & Data Processing

Problem Statement & Objective

Goodreads users need better personalized recommendations to navigate a catalog of 10,000 unique books.

Goal: Use collaborative filtering for initial matching, then an LLM to re-rank and personalize results based on user preferences.

The Datasets

Books: 16 data points (title, author, year, averaging rating, rating count, etc.) for nearly 10,000 unique titles

Ratings: ~165,000 book ratings across 1,192 unique users

Preprocessing

- Excluded books with <10 ratings for higher signal
- 80/20 train/test split
- Set random seed for reproducibility

EDA Highlights

Rating Volume

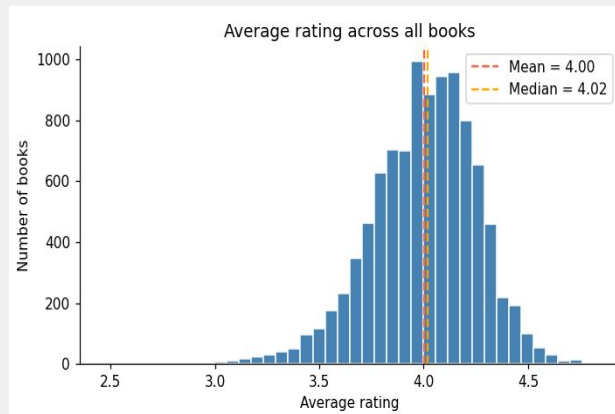
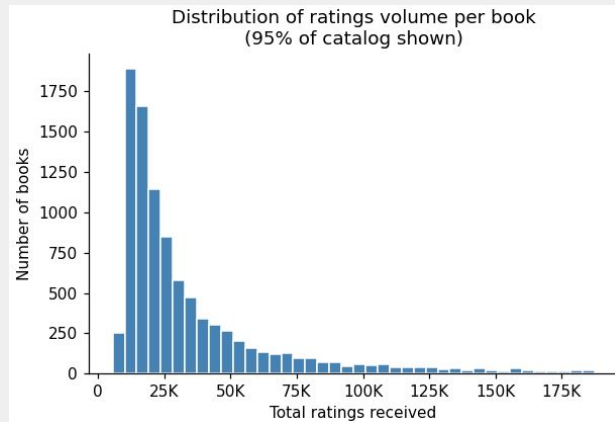
Rating volume per book is **right-skewed**. Most books have relatively few ratings, while a few titles dominate the catalog.

Average Rating

Ratings are approximately **normally distributed**, with a mean and median of **~4 stars**. **Positive selection bias** applies here: readers pick books they expect to enjoy.

Goodreads Catalog Trends

80% of books are in English. Catalog skews **recent** (majority of books are post-1980 publications).



Collaborative Filtering: Modeling Approach

Approach

Compared **three models** (Popularity Baseline, UBCF, IBCF) using default hyperparameters.

Ran **hyperparameter search** to optimize similarity metrics and k-values before final conclusion.

Evaluation Strategy

Primary Metrics:

Precision@10 and Recall@10

Why Not RMSE?

Ranking quality matters more than point prediction accuracy for a recommender app.

Hyperparameter Grid

Similarity:

Pearson, Pearson Baseline, Cosine

Neighborhood size (k):

10, 20, 30, 50

Min neighbors (min_k):

Set at 2 (not tuned)



Context Matters: Being off on a star-rating prediction matters less than leaving a well-matched book out of a user's Top 10 list entirely.

Model Selection & Interpretation

Model	Similarity	k	RMSE ↓	Precision@10 ↑	Recall@10 ↑
Baseline	—	—	0.8468	0.7195	0.5247
UBCF (default)	Pearson	20	1.0220	0.7264	0.5315
IBCF (default)	Cosine	20	0.8994	0.6378	0.4573
UBCF (tuned)	Pearson Baseline	50	0.9670	0.7393	0.5416
★ IBCF (tuned)	Pearson Baseline	20	0.8281	0.7423	0.5467

Performance

Tuning meaningfully improved both CF models.

After tuning, IBCF edges out UBCF and the baseline model across all metrics.

Bias Correction

Tuned IBCF beats the baseline on RMSE by correcting for user and item rating bias before computing similarity.

Explainability

Transparent reasoning builds trust.

IBCF's "Because you liked X" approach is more intuitive than UBCF's "Readers like you liked Y."

LLM Re-Ranking & Prompt Strategy

How It Works:

- 1. IBCF Pool:** Tuned IBCF generates a ranked candidate pool
- 2. LLM Handover:** Candidates pass to Gemini with user preferences.
- 3. Re-Ranking:** Gemini re-ranks and explains each pick.

Structured Schema

Enforces consistent and parseable title, author, and reason fields for every pick.

System Instruction

Restricts model to the candidate list and sets the voice of the explanations.

Temperature

Balances consistency and creative risk.

Low = consistent
High = unexpected

Pool Size

Larger pools give high-temp modes more room to diverge from obvious top picks.

See Appendix A for prompt design detail and Appendix B for output comparisons across modes.

Re-Ranking Modes & Key Findings

Three modes were developed to show how prompt design can enhance the user experience beyond basic re-ranking:

Match Me (temp=0.2)

Focused, on-target picks for exactly what you asked

Roast Me (temp=0.9)

Aligned picks delivered with a comedic voice

Surprise Me (temp=1.4)

Unexpected picks that still fit the mood

Key Findings:

- Different preference types (*mood, taste-matching, situational, content boundaries*) drive meaningful re-ordering that CF cannot accomplish.
- System instruction shapes tone without meaningfully changing which books get recommended.
- Match Me and Roast Me differ in tone, but produce similar picks. Surprise Me surfaces unexpected titles, but explanations still tie back to the user's stated preference.

Business Implications & Lessons Learned

Business Applications

Strong recommendation engines reduce decision fatigue and keep users engaged.

CF provides the personalization at scale, and the LLM makes it interactive.

Together, they create a memorable, high-utility user experience.

Challenges

Cold Start: New users need a fallback like popularity baseline or quiz.

Latency & Cost: LLM calls add cost; trigger only on explicit user input.

Trust & Explainability: Requires monitoring to prevent inaccurate justifications.

Lessons Learned

Ranking metrics (precision and recall) matter more than RMSE for evaluation.

Pearson baseline similarity outperforms default metrics in sparse settings.

Prompt design alone can differentiate the product and drive retention.

Appendix A: LLM Layer Prompt Design & Structure

Structured output fields

`title: str` Exact book title from the candidate list

`author: str` Author name as provided in the candidate list

`reason: str` A brief explanation of why this book fits the reader's preference

Enforced on every response — output is always clean and parseable regardless of mode or temperature.

System instructions & inference parameters by mode

Mode	Temp.	Seed	System instruction
Match Me	0.2	6604	"You are a personal book concierge. Re-rank candidates based on how well they match the reader's stated preference. Use exact titles and authors from the list. Do not suggest books outside the provided list."
Roast Me	0.9	None	"You are a devastatingly sarcastic book critic. Roast their taste mercilessly – be funny, be mean in a loving way. Then give genuinely good recommendations. Each reason should start with the roast and end with why the book is perfect. Think Gordon Ramsay, but for books."
Surprise Me	1.4	None	"You are a contrarian recommender with a talent for hidden gems. Do NOT pick the most obvious titles. Find the unexpected match – the book that fits the request in a surprising or non-obvious way. Be bold. Explain why this overlooked pick is actually perfect."

Model: gemini-2.5-flash-lite | CF stage fetches 100 candidates for all modes; LLM receives 40 (shuffled). Surprise Me benefits from the wider CF pool via random sampling.

Appendix B: Re-Ranking Output Comparison

Preference: "I just got dumped and need a book that will make me believe in love again" · User 35 · Same candidate pool

Match Me temp=0.2

Focused, on-target picks

1. A Court of Mist and Fury

Sarah J. Maas

This book is a highly-rated fantasy romance that focuses on the development of a strong and loving relationship, which could be exactly what you need to believe in love again.

Also recommended:

2. All the Bright Places
3. Aristotle and Dante
4. Forever
5. Obsidian

Roast Me temp=0.9

Aligned picks, comedic voice

1. A Court of Mist and Fury

Sarah J. Maas

Oh, you got dumped and need to believe in love again? You've probably been mainlining YA fantasy with dubious emotional maturity, so why not dive headfirst into more of the same? It will make you forget your pathetic ex ever existed.

Also recommended:

2. Aristotle and Dante
3. Opal (Lux, #3)
4. Obsidian
5. Kiss of Midnight

Surprise Me temp=1.4

Unexpected but justified picks

1. We Are All Completely Beside Ourselves

new

Karen Joy Fowler

This novel masterfully explores the nature of family and belonging, offering a powerful narrative about finding connection and love in unexpected circumstances — exactly what you need to rebuild your faith in the world.

Also recommended:

2. The Sword of Summer new
3. All the Bright Places
4. Aristotle and Dante
5. The Boys in the Boat new

Match Me vs. Roast Me: Both surface the same top pick. The system instruction changes the voice — not the recommendation.

new Titles unique to Surprise Me. Explanations still tie directly back to the stated preference.